

Lecture Notes II

Course: General Topology

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Topic: New Spaces from Old

Proof 3.2 (Preservation of open and closed subsets in the subspace topology). i) If $U \in \mathcal{T}_S$, then $U = S \cap V$ for some $V \in \mathcal{T}_X$ so that if S is also open, U is open in X . Similarly for closed subsets.
ii) For $U \subseteq S$ we have that $U \in \mathcal{T}_X$ implies that $U = U \cap S$ is a possible way of writing U in such way that $U \in \mathcal{T}_S$ and similarly for closed.

Proof 3.3 (Characteristic property of the Subspace topology). $(\Rightarrow)$ Suppose $\iota_S \circ f$ is continuous. If $U \in \mathcal{T}_S$, then $U = S \cap V$ for some $V \in \mathcal{T}_X$, and notice that

$$\iota_S^{-1}(U) = \{s \in S : \iota_S(s) \in U\} = \{s \in S : s \in V\} = S \cap V$$

because ι_S is defined only on S , and thus

$$f^{-1}(U) = f^{-1}(\iota_S^{-1}(U)) = (\iota_S \circ f)^{-1}(U).$$

$(\Leftarrow)$ If f is continuous then for any $V \in \mathcal{T}_X$ we have

$$(\iota_S \circ f)^{-1}(V) = f^{-1}(S \cap V)$$

so that if $f : Y \rightarrow S$ is continuous, $f^{-1}(S \cap V)$ is open for any V open in X and thus $\iota_S \circ f : Y \rightarrow X$ is continuous.

Proof 3.4 (Properties of the Subspace Topology).

- i) Let $R \subseteq S$ be a subspace of S which in turn is a subspace of X and let $U \subseteq R$. Suppose U is open in the topology that R inherits from S meaning that $U = R \cap G$ for some $G \in \mathcal{T}_S$ so that $G = S \cap V$ for some $V \in \mathcal{T}_X$ and thus $U = (R \cap S) \cap V = R \cap V$ so U is open in the topology it inherits from X . Conversely, if U is open in the topology of R inherited from X , then $U = W \cap R$ for some $W \in \mathcal{T}_X$, then since $R \cap S = R$, we have $U = (W \cap S) \cap R$ so U is the intersection of an open subset of S ($W \cap S$) and R , meaning that U is open in the topology of R inherited from S .
- ii) First, it is clear that any element in $\mathcal{B}_S$ is open in S because each element in $\mathcal{B}$ is open in X . We will show that any element in $\mathcal{T}_S$ satisfies the basis criterion with respect to $\mathcal{B}_S$. Let $U \in \mathcal{T}_S$, then $U = V \cap S$ for some $V \in \mathcal{T}_X$, but since $\mathcal{B}$ is a basis for X , V satisfies the basis criterion wrt $\mathcal{B}$, i.e., given any $p \in V$ there is an element $B \in \mathcal{B}$ for which $p \in B \subseteq V$. Particularly, given any $p \in U$, we have $p \in V$ and $p \in S$ so that the former yields $p \in B \subseteq V$ for some $B \in \mathcal{B}$ and this with $p \in S$ gives $p \in B \cap S \subseteq V \cap S = U$ (where $B \cap S \in \mathcal{B}_S$) and the proof is done.
- iii) Let $p_i, p \in S$. Suppose that $p_i \rightarrow p$ in S , and thus $p_i, p \in X$ so for any open neighbourhood U of p in X we have that $S \cap U$ is a relative neighbourhood of p in S and by our assumption there exists N such that $p_n \in S \cap U$ for any $n \geq N$ but since $p_i \in S$ by hypothesis, this is equivalent to $p_i \in U$ so $p_i \rightarrow p$ in X . Conversely, suppose that $p_i \rightarrow p$ in X and hence take any relative open neighbourhood $S \cap U$ of p for which we must have $p \in U$, and thus we choose N so that $p_n \in U$ for $n \geq N$ and again, since $p_i \in S$ we get that $p_n \in S \cap U$ for $n \geq N$ meaning that $p_i \rightarrow p$ in S .
- iv) Straightforward
- v) By ii) we have that $\mathcal{B}_S = \{S \cap B : B \in \mathcal{B}\}$ is a basis for S provided that $\mathcal{B}$ is a basis for X so that if $\mathcal{B}$ is countable, then so is $\mathcal{B}_S$.

Proof 3.5 (The inclusion map is a topological embedding). Let $\iota_S : S \hookrightarrow X$, then clearly ι_S is injective and seen as a map onto its image $\iota_S : S \rightarrow S$ this is just Id_S which is continuous (in the subspace topology) and its inverse is itself so $\iota_S \in \text{Homeo}(S, S)$ and thus ι_S is a topological embedding.

Proof 3.7 (Continuity+injectivity+open or closed implies embedding). Suppose $f : X \rightarrow Y$ is a continuous injective map. As always, f is bijective when seen as a map $f : X \rightarrow f(X)$ which is still continuous by the corollary preceding Theorem 3.4 (restricting the codomain) and now we only need to show that it is a homeomorphism, we only need to show that it is open or closed as a map to $f(X)$ —a bijective continuous map is a homeomorphism if and only if it is open or closed—

Suppose $f : X \rightarrow Y$ is open, then for any open A in X we have that $f(A)$ is open in Y , and by Theorem 3.2 we have that $f(A)$ is open in $f(X) \subseteq Y$ so f is open as a map $f : X \rightarrow f(X)$ and similarly for “closed”.

Proof 3.8 (Surjective embeddings are homeomorphisms). The conclusion readily follows since surjectivity means that $f : X \rightarrow Y$ is such that $f(X) = Y$ so $f \in \text{Homeo}(X, Y)$.

Proof 3.12 (Gluing lemma). We can always define $f(x) = f_i(x)\chi_{A_i}(x)$ which is well-defined by the assumption of agree on overlaps. If the sets A_i are open then the continuity of f follows directly from Theorem 2.10 (Local criterion for continuity). If the sets A_i form a finite closed cover of X then by Theorem 2.8 we only need to prove that the preimage of closed sets is closed. Notice that $f_i^{-1}(K) = f^{-1}(K) \cap A_i$ and since $\bigcup_i A_i = X$ and $X \cap f^{-1}(K) = f^{-1}(K)$ we have $f^{-1}(K) = \bigcup_{i=1}^n (f^{-1}(K) \cap A_i) = \bigcup_{i=1}^n (f_i^{-1}(K))$ but $f_i^{-1}(K)$ is closed in A_i by continuity of each f_i and also A_i is closed in X so $f_i^{-1}(K)$ is closed in X for each i due to Theorem 3.2. Hence $f^{-1}(K)$ is closed in X provided that K is closed in X (the finite union of closed sets is closed).

Proof 3.13 (Uniqueness of the subspace topology). Suppose there is some topology on S for which the characteristic property holds and let S_g be the subset endowed with such topology, also let S_s be the subset endowed with the subspace topology. We know (see problem sheet) that for S_s, S_g , a bijective map $f : S_s \rightarrow S_g$ is a homeomorphism if and only if $f(\mathcal{T}_s) = \mathcal{T}_g$. We will show that $\text{Id}_{sg} : S_s \rightarrow S_g$ (S_s and S_g are the same sets but with different topologies) is a homeomorphism. First, $\iota_s : S_s \hookrightarrow X$ and $\iota_g : S_g \hookrightarrow X$ are continuous because of the characteristic property. Next, consider the two composite maps

$$\begin{array}{ccc} & X & \\ \iota_g \circ \text{Id}_{sg} \nearrow & \uparrow & \nearrow \iota_s \\ S_s & \xrightarrow{\text{Id}_{sg}} & S_g \end{array} \quad \begin{array}{ccc} & X & \\ \iota_s \circ \text{Id}_{gs} \nearrow & \uparrow & \nearrow \iota_g \\ S_g & \xrightarrow{\text{Id}_{gs}} & S_s \end{array}$$

where both Id_{sg} and Id_{gs} represent the identity map of S (again, they are the same set), and ι_s, ι_g are the inclusion map of S into X (the difference in notation is just to distinguish the sense of continuity). We note that $\iota_g = \iota_s \circ \text{Id}_{gs}$ and $\iota_s = \iota_g \circ \text{Id}_{sg}$, but we showed before that both of these are continuous so that using the characteristic property again we get that both $\text{Id}_{sg}, \text{Id}_{gs}$ are continuous and they are inverse of each other so Id_{sg} is a homeomorphism of S_s onto S_g , and we thus conclude that $\text{Id}_{sg}(\mathcal{T}_s) = \mathcal{T}_s = \mathcal{T}_g$.

Proof 3.14 (The product open sets form a basis). Let $X = \prod X_i$ and let

$$\mathcal{B} = \left\{ \prod_{i=1}^n U_i : U_i \in \mathcal{T}_i \ i = 1, \dots, n \right\}$$

then i) given that $X_i \in \mathcal{T}_i$, $X = \prod X_i \in \mathcal{B}$ so $X \subseteq \bigcup \mathcal{B}$ and given that $\mathcal{T}_i \subseteq X_i$ we clearly also have $\bigcup \mathcal{B} \subseteq X$. ii) Take $P_1, P_2 \in \mathcal{B}$ and let $x \in P_1 \cap P_2$ so $x \in P_1$ and hence $x^i \equiv \pi^i(x) \in \pi^i(P_1) \equiv P_1^i$ for any $i = 1, \dots, n$, but since each P_1^i is open, there is an open neighbourhood V_1^i of x^i contained in P_1^i and a similar reasoning applies to P_2 so we get an open neighbourhood V_2^i of x^i contained in P_2^i , we thus have that $V_1^i \cap V_2^i$ is an open neighbourhood of x^i contained in $P_1^i \cap P_2^i$ and hence let $P_3 = \prod_i V_1^i \cap V_2^i$ so that clearly $x \in P_3 \subseteq P_1 \cap P_2$.

Proof 3.15 (Characteristic property of the product topology). Suppose f_i is continuous. To prove

that f is continuous, take $\prod_i U_i$ to be any element of the basis and thus

$$f^{-1}\left(\prod_{i=1}^n U_i\right) = \bigcap_{i=1}^n f_i^{-1}(U_i)$$

so the preimage of any element in the basis is open and hence f is continuous by Theorem 2.19. Conversely, if $f : Y \rightarrow \prod_i X_i$ is continuous, then the composition $\pi^i \circ f = f_i$ is also continuous because for any open U_i in X_i we have that $X_1 \times X_2 \times \cdots \times U_i \times \cdots \times X_n$ is open and by continuity of f , its preimage

$$f^{-1}(X_1 \times X_2 \times \cdots \times U_i \times \cdots \times X_n) = f_1^{-1}(X_1) \cap \cdots \cap f_i^{-1}(U_i) \cap \cdots \cap f_n^{-1}(X_n) = f_i^{-1}(U)$$

is open in Y and the proof is complete.

Corollary (canonical projections are always continuous). Apply the characteristic property to $Y = \prod_i X_i$ and $f = \text{Id}_Y$ (the identity map is always continuous).

Proof 3.17 (Properties of the product topology)

Proof 3.18 (Product map preserves continuity and homeomorphisms). Continuity of a map can be fully characterised by a basis, i.e., if the preimage of any basis subset of X is open, then the map is continuous on X , so let $f_i : X_i \rightarrow Y_i$ and define the product map

$$\prod_j f_j(x) = (f_1(x^1), \dots, f_n(x^n))$$

then notice how the preimage of an element of a basis in Y

$$\left(\prod_j f_j\right)^{-1}(U_1 \times \cdots \times U_n) = \prod_j f_j^{-1}(U_j), \quad U_j \in \mathcal{T}_{Y,j}$$

is the product of open subsets $f_j^{-1}(U_j)$ —each of which is open in X_j — so it is open in $\prod_j X_j$. Moreover, the inverse map of a product map is another product map so the product of homeomorphisms is a homeomorphism.

Proof 3.34 (Subgroups and direct products of topological groups are topological groups). Let G be a topological group and let $K \subseteq G$ be a subgroup of G . Then equip K with the subspace topology and thus, given that the maps $\star : G \times G \rightarrow G$ and $i : G \rightarrow G$ are continuous, we can restrict the domain to the subspaces $K \times K$ of $G \times G$ and K of G so $\star : K \times K \rightarrow G$ and $i : K \rightarrow G$ are both continuous, moreover, given that K is a subgroup, the maps are also closed, meaning that $\star(K \times K) \subseteq K$ and $i(K) \subseteq K$ and thus we can also restrict the codomain to get that $\star : K \times K \rightarrow K$ and $i : K \rightarrow K$ are both continuous. For the case on direct products, let G, H be two topological groups and equip $G \times H$ with the product $\star$ defined via

$$(g, h) \star (g', h') = (g \star_G g', h \star_H h')$$

and also with the product topology, so that clearly, the canonical projections are $\pi^G \circ \star = \star_G, \pi^H \circ \star = \star_H$ and by the characteristic property of the product topology, $\star$ is continuous, similarly for i defined by

$$i(g, h) = (i_G(g), i_H(h))$$

which is the product of the continuous maps i_G, i_H defined via $i_G(g) = g^{-1}, i_H(h) = h^{-1}$ for all $g \in G, h \in H$ and thus continuous itself.

Corollary. Using the previous result, it is easy to see that the circle $\mathbb{S}^1$, which is a subgroup of $\mathbb{C}^* = \mathbb{C} \setminus \{0\}$ under the usual multiplication, is itself a topological group and thus the n -torus $\mathbb{T}^n = \mathbb{S}^1 \times \cdots \times \mathbb{S}^1$ is also a topological group.

Proof 3.36 (Relation between continuous actions and actions by homeomorphisms). First suppose that the action $G \times X \rightarrow X$ of G on X is continuous, in particular, by restricting the domain, we have that $L_g^A : X \rightarrow X$ given by $L_g^A(x) = g \cdot x$ is continuous for any given $g \in G$, and the definition of being a group action guarantees that it has an inverse $x \mapsto g^{-1} \cdot x$ which is clearly also continuous ($g^{-1} \in G$). We thus conclude that $x \mapsto g \cdot x$ is a homeomorphism and thus G acts by homeomorphisms. Now suppose G is a discrete group. If G acts by homeomorphisms, then the map $G \times X \rightarrow X$ is continuous when restricted to $\{g\} \times X$ so that each point in $G \times X$ has an open neighbourhood on which the restriction of the action is continuous, meaning that the action is continuous by Theorem 2.10.